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A generalized gamma kernel-machine promotion-time cure model

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ABSTRACT

Cure models estimate the fraction of individuals who never experience the event of interest, but linear promotion-time predictors can misrepresent nonlinear covariate effects on the latent-cause mechanism. The generalized gamma kernel-machine promotion-time cure model places the log latent-cause intensity in a reproducing-kernel Hilbert space with an explicit unpenalized intercept, retains a common Stacy generalized gamma latency, and accommodates Poisson and negative-binomial latent counts. Estimation uses safeguarded penalized expectation-maximization or expectation/conditional maximization either, followed by bounded observed-likelihood refinement and explicit convergence diagnostics. Prediction for unseen covariates uses training-only preprocessing and the fitted kernel expansion. Evaluation uses common event-stratified 70/30 splits, inner tuning confined to training data, inverse-probability-of-censoring weighted Brier scores, paired uncertainty, retained failures, and versioned simulation and melanoma outputs. The comparison includes log-linear, additive spline, quadratic, and radial-basis kernel predictors under both count laws, together with separate primary and mechanism designs. Quantitative conclusions are restricted to reproduced outputs from the declared implementation and do not assert universal superiority, causal effects, or identification of individual cure status.

1 | Introduction

Standard survival models assign vanishing probability to remaining event free as follow-up grows arbitrarily long. That limit is inappropriate when a clinically meaningful subgroup will never experience the event of interest, a situation signalled by a sustained plateau in the survival curve under mature follow-up. Cure models describe this setting by separating the long-term event-free probability from the timing of events among individuals who remain susceptible, which makes the cured proportion an interpretable quantity in oncology and other long-term-survivor studies.[1,2]

Two broad constructions are common. Mixture cure models specify the cured proportion and the survival of susceptible individuals as separate components. Non-mixture models embed the nonzero limiting survival probability in a population survival law, with the promotion-time model as a principal example.[3] The promotion-time mechanism has a two-stage interpretation. An initiation stage determines the unobserved number N_i of latent causes carried by an individual; a maturation stage assigns an activation time to each cause, and the first activation produces the event. No activation is possible when $N_i = 0$, which identifies the cure mass. Writing the count law through its probability-generating function maps the latent activation-time distribution to both population survival and the cure fraction.

The expected latent count is usually modelled as a log-linear function of covariates. This choice is convenient but can conceal threshold, interaction, and periodic associations between covariates and long-term event-free probability. The activation-time component carries a parallel risk, because an overly restrictive latency distribution can transfer misspecification to the estimated covariate effect and to the cure fraction. These concerns motivate

joint consideration of a flexible latent-intensity predictor and a sufficiently general parametric latency.

Flexible cure-model predictors have been developed with smoothers, support vector machines, neural networks, and deep architectures, so nonlinear and kernel-based cure modelling is an established and active field rather than a new idea. The support-vector promotion-time model of Pal and Aselisewine allows a nonlinear incidence component, and Xie and Yu use a neural network for an unstructured promotion-time component.[4,5] Deep promotion-time models and structured nonlinear mixture-cure models further broaden the available high-dimensional architectures.[6,7] However, these approaches lack a unified likelihood-based framework that simultaneously provides a flexible parametric latency, a transparent count-family construction for overdispersion, and fully traceable leakage-free evaluation.

This article develops a specific combination of an RKHS latent intensity, a generalized gamma latency, and Poisson and negative-binomial counts, with penalized EM and ECME estimation, bounded observed-likelihood refinement, and traceable leakage-free evaluation.

2 | Methods

The GG-KM model combines a promotion-time cure mechanism with a flexible latency distribution and a regularized kernel predictor for the expected number of latent causes. This section defines the observed data, the count-family construction, the Poisson and negative-binomial specifications, and their penalized estimation procedures. The observed likelihoods, conditional objective blocks, and estimation algorithms are stated here; lengthy likelihood derivations and analytical gradients are retained in the Supporting Information.

Figure 1 lays out the GG-KM model and its estimation and evaluation path. Panel A shows the latent-cause intensity $\theta(\boldsymbol{x}) = \exp\{f(\boldsymbol{x})\}$ built by a radial-basis kernel expansion over $(x_1, x_2) \in [-1, 1]^2$, with the six kernel centres $\boldsymbol{x}_j$ marked and the count law named beneath the surface. Panel B plots four parallel time axes for two cured individuals ($N_i = 0$) and two susceptible ones ($N_i \geq 1$). Its key maps every glyph to a quantity: an open circle is an activation time Z_{ij} from the generalized gamma latency, a filled circle is the event time $T_i = \min_j Z_{ij}$, and a vertical bar is the censoring time C_i; the resulting pair (Y_i, δ_i) is printed at the right of each axis. Panel C draws the matched Poisson and negative-binomial population survival curves, the cure plateau $p_c(\boldsymbol{x})$, and the shaded interval at or below the scoring horizon τ. Panel D traces the leakage-controlled pipeline: event-stratified 70/30 split, Gram matrix construction, penalized EM or ECME with bounded refinement, kernel-coefficient prediction, and IPCW Brier evaluation. The dashed arc in Panel D is the only feedback path and marks the single iterative update that revises the kernel coefficients, the latency parameters, and the negative-binomial dispersion when applicable.

2.1 | Promotion-time cure model

For individual i, let T_i denote the event time and C_i an independent right-censoring time conditional on the modeled covariates. The observed data are $Y_i = \min(T_i, C_i)$ and $\delta_i = I(T_i \leq C_i)$, together with a covariate vector $\boldsymbol{x}_i$. The latent variable N_i counts potential causes of the event, and $N_i = 0$ identifies a cured individual under the promotion-time construction.

The promotion-time mechanism links the latent-cause count to the observed time in two stages. Conditional on the covariates, $N_i \mid \boldsymbol{x}_i$ has probability mass function $p_n(\boldsymbol{x}_i) = P(N_i = n \mid \boldsymbol{x}_i)$ and probability-generating function $A_i(u) = E(u^{N_i} \mid \boldsymbol{x}_i) = \sum_{n\geq 0} p_n(\boldsymbol{x}_i)u^n$. Given $N_i = n \geq 1$, the n causes activate at independent and identically distributed times with proper survival function $S_{GG} = 1 - F_{GG}$ that is independent of N_i, and the event time is the earliest activation, $T_i = \min(Z_{i1}, \ldots, Z_{iN_i})$; when $N_i = 0$ no cause activates and $T_i = \infty$.

Summing over the latent counts, the population survival function and the cure fraction follow from the generating function as

$$\begin{aligned} S_{\text{pop}}(t \mid \boldsymbol{x}_i) &= P(T_i > t \mid \boldsymbol{x}_i) \\ &= \sum_{n\geq 0} p_n(\boldsymbol{x}_i) S_{GG}(t)^n = A_i\{S_{GG}(t)\}, \\ p_c(\boldsymbol{x}_i) &= \lim_{t\to\infty} S_{\text{pop}}(t \mid \boldsymbol{x}_i) \\ &= A_i(0) = P(N_i = 0 \mid \boldsymbol{x}_i). \end{aligned} \tag{1}$$

The population survival is therefore the generating function evaluated at the latency survival, and the cure fraction is its value at zero, equal to the probability of no latent cause. Differentiating the population survival with respect to time, and using $S'_{GG} = -f_{GG}$, gives the population density and hazard,

$$\begin{aligned} f_{\text{pop}}(t \mid \boldsymbol{x}_i) &= f_{GG}(t)\, A'_i\{S_{GG}(t)\}, \\ h_{\text{pop}}(t \mid \boldsymbol{x}_i) &= \frac{f_{\text{pop}}(t \mid \boldsymbol{x}_i)}{S_{\text{pop}}(t \mid \boldsymbol{x}_i)}. \end{aligned} \tag{2}$$

The density weights the latency density by the derivative of the generating function at the latency survival, and the hazard is their ratio. These quantities satisfy the required boundary behavior: $S_{\text{pop}}(0 \mid \boldsymbol{x}) = 1$, $S_{\text{pop}}(t \mid \boldsymbol{x})$ is non-increasing in t, and $\lim_{t\to\infty} S_{\text{pop}}(t \mid \boldsymbol{x}) = p_c(\boldsymbol{x})$. In the limiting case $p_c(\boldsymbol{x}) = 1$ the latent-count mass concentrates on $N_i = 0$, so $S_{\text{pop}}(t \mid \boldsymbol{x}) = 1$ for all t. The generic population survival S_{pop} and cure fraction p_c are specialized to the Poisson and negative-binomial count laws in the next subsection.

2.2 | Count-family unification

The count probability-generating function provides a common representation of mixture and promotion-time cure models.[1,3,2] Let $p_s(\boldsymbol{x}) = 1 - p_c(\boldsymbol{x})$ denote the susceptible probability. A Bernoulli count $N_i \in \{0, 1\}$ produces the classical two-component mixture model, whereas Poisson and negative-binomial counts produce non-mixture promotion-time models. Their generating

[Figure 1 near here]

FIGURE 1 | Structure of the GG-KM model and of its estimation and evaluation. Panel A shows the generative mechanism: a kernel predictor $f(\boldsymbol{x}) = \beta_0 + \sum_j \alpha_j K(\boldsymbol{x}, \boldsymbol{x}_j)$ over the marked kernel centres $\boldsymbol{x}_j$ determines the positive latent-cause intensity $\theta(\boldsymbol{x}) = \exp\{f(\boldsymbol{x})\}$, which indexes the Poisson or negative-binomial count N_i named beneath the surface. Panel B shows the observation mechanism. Given $N_i \geq 1$ the promotion times Z_{ij} are independent generalized gamma variables and $T_i = \min_j Z_{ij}$, whereas $N_i = 0$ gives $T_i = \infty$, so only $Y_i = \min(T_i, C_i)$, $\delta_i = I(T_i \leq C_i)$ and $\boldsymbol{x}_i$ are observed; the key maps the open circle, the filled circle and the vertical bar to Z_{ij}, T_i and C_i, and the realized pair (Y_i, δ_i) is printed at the right of each time axis. Panel C shows the population survival curves, where P denotes Poisson and NB negative-binomial counts. They use the simulation latency $(a, d, p) = (1.6, 1.3, 1.1)$, a matched cure fraction of 0.45 and dispersion $\phi = 1$; they are illustrative rather than fitted, and line style with markers distinguishes them in addition to color. Shading marks times at or below the frozen horizon τ within which predictions are scored, so the approach of $S_{\text{pop}}(t \mid \boldsymbol{x})$ to $p_c(\boldsymbol{x})$ is extrapolated rather than evaluated. Panel D shows the estimation and evaluation path: one event-stratified 70/30 split with training-only preprocessing, the validated Gram matrix, penalized EM or ECME with an ascent safeguard, out-of-sample prediction, and IPCW integrated Brier scoring with paired differences. The dashed arc in Panel D is the only feedback path and marks the iterative update of $(\beta_0, \boldsymbol{\alpha})$, (a, d, p) and ϕ.

functions, population survival functions, and cure fractions are

$$\begin{aligned}
A_{\mathrm{B},i}(u) &= 1 - p_s(\boldsymbol{x}_i) + p_s(\boldsymbol{x}_i)u, \\
S_{\mathrm{B}}(t \mid \boldsymbol{x}_i) &= 1 - p_s(\boldsymbol{x}_i) + p_s(\boldsymbol{x}_i)S_{GG}(t), \\
p_{c,\mathrm{B}}(\boldsymbol{x}_i) &= 1 - p_s(\boldsymbol{x}_i), \\
A_{\mathrm{P},i}(u) &= \exp\{\theta_i(u-1)\}, \\
S_{\mathrm{P}}(t \mid \boldsymbol{x}_i) &= \exp\{-\theta_i F_{GG}(t)\}, \\
p_{c,\mathrm{P}}(\boldsymbol{x}_i) &= \exp(-\theta_i), \\
A_{\mathrm{NB},i}(u) &= \{1 + \phi\theta_i(1-u)\}^{-1/\phi}, \\
S_{\mathrm{NB}}(t \mid \boldsymbol{x}_i) &= \{1 + \phi\theta_i F_{GG}(t)\}^{-1/\phi}, \\
p_{c,\mathrm{NB}}(\boldsymbol{x}_i) &= \{1 + \phi\theta_i\}^{-1/\phi}.
\end{aligned} \tag{3}$$

The Bernoulli construction separates the cured mass from the susceptible latency survival. The Poisson and negative-binomial constructions instead allow any nonnegative number of latent causes and identify cure with the zero-count probability. The negative-binomial family approaches the Poisson family as $\phi \to 0$. Its probability-generating function also yields the Bernoulli form under the algebraic continuation $\phi = -1$ and $\theta_i = p_s(\boldsymbol{x}_i) \in [0, 1]$. That continuation explains the connection between mixture and promotion-time formulations, but it is outside the fitted GG-KM parameter space, which restricts the negative-binomial dispersion to $\phi > 0$ and the intensity to $\theta(\boldsymbol{x}) = \exp\{f(\boldsymbol{x})\}$.

The Bernoulli case recovers the standard mixture cure model, in which the cured mass $1 - p_s(\boldsymbol{x})$ is separated from the susceptible latency survival S_{GG}. No Bernoulli or binomial estimator is introduced here; the active GG-KM models use only Poisson and negative-binomial latent counts.

2.3 | Generalized gamma kernel-machine model

The activation time of each latent cause follows the Stacy generalized gamma distribution with scale $a > 0$ and shape parameters $d > 0$ and $p > 0$.[8] Its density, distribution function, and survival function are

$$\begin{aligned}
f_{GG}(t; a, d, p) &= \frac{p t^{d-1} \exp\{-(t/a)^p\}}{a^d \Gamma(d/p)}, \\
F_{GG}(t; a, d, p) &= P\{d/p, (t/a)^p\}, \\
S_{GG}(t; a, d, p) &= Q\{d/p, (t/a)^p\},
\end{aligned} \tag{4}$$

where P and Q are the regularized lower and upper incomplete gamma functions. This parameterization contains Weibull models when $d = p$, gamma models when $p = 1$, and the exponential model when $d = p = 1$. The lognormal distribution is not a special case or limit of the Stacy parameterization at fixed (a, d, p); obtaining it requires a separate reparameterized limit and is therefore not claimed here. The implementation verifies numerically that the density integrates to one and that the derivative of the distribution function equals the density, so F_{GG}, f_{GG}, and S_{GG} remain mutually consistent. Figure 2 illustrates how the verified special cases and a general parameter setting produce distinct density and hazard shapes.

The latent-cause intensity depends on the covariates through a kernel predictor whose reproducing kernel is chosen from a fixed family. For inputs $\boldsymbol{x}$ and $\boldsymbol{z}$, the candidate kernels are

$$\begin{aligned}
K_{\text{lin}}(\boldsymbol{x}, \boldsymbol{z}) &= \boldsymbol{x}^\top \boldsymbol{z}, \\
K_{\text{poly}}(\boldsymbol{x}, \boldsymbol{z}) &= (\gamma\, \boldsymbol{x}^\top \boldsymbol{z} + c_0)^q, \\
K_{\text{rbf}}(\boldsymbol{x}, \boldsymbol{z}) &= \exp\{-\gamma \|\boldsymbol{x} - \boldsymbol{z}\|_2^2\}, \\
K_{\text{exp}}(\boldsymbol{x}, \boldsymbol{z}) &= \exp\{-\gamma \|\boldsymbol{x} - \boldsymbol{z}\|_2\}, \\
K_{\text{lap}}(\boldsymbol{x}, \boldsymbol{z}) &= \exp\{-\gamma \|\boldsymbol{x} - \boldsymbol{z}\|_1\}, \\
K_{\text{cau}}(\boldsymbol{x}, \boldsymbol{z}) &= \{1 + \gamma \|\boldsymbol{x} - \boldsymbol{z}\|_2^2\}^{-1}, \\
K_{\text{per}}(\boldsymbol{x}, \boldsymbol{z}) &= \exp\Big\{-\tfrac{2}{\ell^2} \textstyle\sum_j \sin^2 \frac{\pi(x_j - z_j)}{\rho_j}\Big\}, \\
K^{\dagger}_{\text{sig}}(\boldsymbol{x}, \boldsymbol{z}) &= \tanh(\gamma\, \boldsymbol{x}^\top \boldsymbol{z} + c_0).
\end{aligned} \tag{5}$$

Here $\gamma > 0$ is a scale parameter, $q \geq 1$ is an integer polynomial degree, $c_0 \geq 0$ is an offset, $\ell > 0$ is a length scale, and each $\rho_j > 0$ is a coordinate period. The exponential kernel uses the Euclidean (L_2) norm, whereas the Laplacian kernel uses the L_1 norm, so the two are distinct. Gaussian and radial-basis denote the single kernel K_{rbf}, which is not duplicated. The dagger marks the conditionally admissible sigmoid candidate: it is not positive semidefinite in general and is retained only when its training Gram matrix passes the declared symmetry and positive-semidefinite gate independently in every tuning fold. A rejected sigmoid candidate is recorded as a tuning failure rather than repaired silently.

Each admissible positive-semidefinite kernel induces a reproducing-kernel Hilbert space $\mathcal{H}_K$. The representer theorem can be stated for $g \in \mathcal{H}_K$ through an objective of the form

$$\begin{aligned}
&\mathcal{J}(g) = \mathcal{L}\{g(\boldsymbol{x}_1), \ldots, g(\boldsymbol{x}_n)\} + G(\|g\|_{\mathcal{H}_K}), \\
&G \text{ strictly increasing} \quad \Rightarrow \quad g^\star(\cdot) = \sum_{j=1}^{n} \alpha_j K(\cdot, \boldsymbol{x}_j).
\end{aligned} \tag{6}$$

To see the finite-dimensional reduction, decompose g into its projection onto the span of $K(\cdot, \boldsymbol{x}_1), \ldots, K(\cdot, \boldsymbol{x}_n)$ and an orthogonal remainder. The remainder vanishes at every training point, leaves

[Figure 2 near here]

FIGURE 2 | Generalized gamma latency flexibility under the Stacy parameterization. Panel A shows the density $f_{GG}(t)$ and Panel B the corresponding hazard $h_{GG}(t) = f_{GG}(t)/S_{GG}(t)$ for the exponential case $(a, d, p) = (1.6, 1, 1)$, Weibull case $(1.6, 0.7, 0.7)$, gamma case $(1.6, 2, 1)$, and the general generalized gamma setting $(1.6, 1.3, 1.1)$ used in the simulation study. Line type and markers distinguish the four cases in addition to color.

$\mathcal{L}$ unchanged, and can only increase the norm, so it cannot occur in a minimizer when G is strictly increasing.[9,10] The theorem supplies the kernel expansion; GG-KM then adds the explicit unpenalized intercept β_0 as a modeling choice. The fitted predictor is therefore

$$f(\boldsymbol{x}) = \beta_0 + \sum_{j=1}^{n} \alpha_j K(\boldsymbol{x}, \boldsymbol{x}_j), \tag{7}$$

where β_0 is the unpenalized intercept and K is a positive-semidefinite kernel. The penalty is $\lambda\boldsymbol{\alpha}^\top \boldsymbol{K}\boldsymbol{\alpha}/2$, where $\lambda \geq 0$ and $\boldsymbol{K}$ is the training Gram matrix. For $\lambda > 0$, the quadratic penalty supplies the strictly increasing RKHS-norm term in Equation (6); when $\lambda = 0$, the implementation selects the same finite-span representative without claiming uniqueness outside that span. The positive latent-cause intensity is $\theta(\boldsymbol{x}) = \exp\{f(\boldsymbol{x})\}$, so $f(\boldsymbol{x}) = \log\theta(\boldsymbol{x})$ is real-valued and the intensity remains positive without an additional constraint.

The count law converts the intensity into population survival and a cure fraction through its generating function. Writing $\theta_i = \theta(\boldsymbol{x}_i)$, Poisson latent counts have generating function

$$A_i(u) = \exp\{\theta_i(u-1)\}, \tag{8}$$

which is the generating function of a Poisson variable with mean θ_i. Substituting $u = S_{GG}(t)$ into Equation (1) gives the population survival, and $u = 0$ gives the cure fraction,

$$\begin{aligned} S_{\mathrm{P}}(t \mid \boldsymbol{x}) &= \exp\{-\theta(\boldsymbol{x})F_{GG}(t)\}, \\ p_{c,\mathrm{P}}(\boldsymbol{x}) &= \exp\{-\theta(\boldsymbol{x})\}. \end{aligned} \tag{9}$$

The corresponding population density and hazard are

$$\begin{aligned} f_{\mathrm{pop}}(t \mid \boldsymbol{x}) &= \theta(\boldsymbol{x}) f_{GG}(t) \exp\{-\theta(\boldsymbol{x})F_{GG}(t)\}, \\ h_{\mathrm{pop}}(t \mid \boldsymbol{x}) &= \theta(\boldsymbol{x}) f_{GG}(t), \end{aligned} \tag{10}$$

so the Poisson population hazard is the latency density scaled by the intensity. Negative-binomial latent counts with mean θ_i and dispersion $\phi > 0$ have generating function

$$A_i(u) = \{1 + \phi\theta_i(1-u)\}^{-1/\phi}, \tag{11}$$

with mean $E(N_i \mid \boldsymbol{x}_i) = \theta_i$ and variance $\mathrm{Var}(N_i \mid \boldsymbol{x}_i) = \theta_i + \phi\theta_i^2$, so ϕ controls overdispersion relative to the Poisson case. The same substitutions give

$$\begin{aligned} S_{\mathrm{NB}}(t \mid \boldsymbol{x}) &= \{1 + \phi\theta(\boldsymbol{x})F_{GG}(t)\}^{-1/\phi}, \\ p_{c,\mathrm{NB}}(\boldsymbol{x}) &= \{1 + \phi\theta(\boldsymbol{x})\}^{-1/\phi}, \end{aligned} \tag{12}$$

and the population density and hazard are

$$\begin{aligned} f_{\mathrm{pop}}(t \mid \boldsymbol{x}) &= \theta(\boldsymbol{x}) f_{GG}(t) \{1 + \phi\theta(\boldsymbol{x})F_{GG}(t)\}^{-1/\phi - 1}, \\ h_{\mathrm{pop}}(t \mid \boldsymbol{x}) &= \frac{\theta(\boldsymbol{x}) f_{GG}(t)}{1 + \phi\theta(\boldsymbol{x})F_{GG}(t)}. \end{aligned} \tag{13}$$

The generic population survival S_{pop} and cure fraction p_c thus specialize to the Poisson forms S_{P} and $p_{c,\mathrm{P}}$ and the negative-binomial forms S_{NB} and $p_{c,\mathrm{NB}}$. The negative-binomial formulation approaches the Poisson formulation as ϕ tends to zero, which provides a direct numerical and interpretive check on the two count models.

2.4 | Penalized estimation

Estimation maximizes the observed penalized log-likelihood of each count model, which combines the kernel predictor, the generalized gamma latency, and the penalty defined above. For the Poisson model, the observed penalized log-likelihood is

$$\ell_{\mathrm{P}}^{\mathrm{pen}} = \sum_{i=1}^{n} [\delta_i\{f_i + \log f_{GG}(Y_i)\} - \theta_i F_{GG}(Y_i)] - \frac{\lambda}{2}\boldsymbol{\alpha}^\top \boldsymbol{K}\boldsymbol{\alpha}, \tag{14}$$

where $f_i = f(\boldsymbol{x}_i)$ and $\theta_i = \exp(f_i)$. For the negative-binomial model, the observed penalized log-likelihood becomes

$$\begin{aligned} \ell_{\mathrm{NB}}^{\mathrm{pen}} = \sum_{i=1}^{n} \Big[&\delta_i\{f_i + \log f_{GG}(Y_i)\} \\ &- \left(\frac{1}{\phi} + \delta_i\right) \log\{1 + \phi\theta_i F_{GG}(Y_i)\} \Big] - \frac{\lambda}{2}\boldsymbol{\alpha}^\top \boldsymbol{K}\boldsymbol{\alpha}. \end{aligned} \tag{15}$$

Estimation builds on the complete-data log-likelihood that conditions on the latent counts. For the Poisson model, up to an additive constant that does not depend on the parameters, it is

$$\begin{aligned} \ell_c = &\sum_{i=1}^{n} \{N_i f_i - \theta_i\} \\ &+ \sum_{i=1}^{n} \{\delta_i \log f_{GG}(Y_i) \\ &\quad + (N_i - \delta_i) \log S_{GG}(Y_i)\}. \end{aligned} \tag{16}$$

Taking the conditional expectation of Equation (16) given the observed data yields an auxiliary objective that separates into a predictor block and a generalized gamma latency block. With the current conditional counts held fixed, the Poisson predictor block is

$$Q_{f,\mathrm{P}} = \sum_{i=1}^{n} \{\widehat{N}_i f_i - \exp(f_i)\} - \frac{\lambda}{2}\boldsymbol{\alpha}^\top \boldsymbol{K}\boldsymbol{\alpha}. \tag{17}$$

For negative-binomial counts, holding the current dispersion ϕ fixed gives

$$Q_{f,\mathrm{NB}} = \sum_{i=1}^{n} \left[\widehat{N}_i f_i - \left(\widehat{N}_i + \frac{1}{\phi}\right) \log\{1 + \phi \exp(f_i)\}\right] - \frac{\lambda}{2}\boldsymbol{\alpha}^\top \boldsymbol{K}\boldsymbol{\alpha}. \tag{18}$$

Both count models share the latency block

$$Q_{GG} = \sum_{i=1}^{n} \{\delta_i \log f_{GG}(Y_i) + (\widehat{N}_i - \delta_i) \log S_{GG}(Y_i)\}. \tag{19}$$

Thus $(\beta_0, \boldsymbol{\alpha})$ occurs only in the relevant predictor block and (a, d, p) only in Q_{GG} once $\widehat{N}_i$ is fixed. The Supporting Information derives these blocks from the complete-data construction and gives their analytical gradients.

The Poisson model is fitted by penalized expectation-maximization.[11] At iteration k, the conditional latent count is

$$\widehat{N}_i^{(k+1)} = \delta_i + \theta_i^{(k)} S_{GG}^{(k)}(Y_i). \tag{20}$$

The conditional maximization then updates $(\beta_0, \boldsymbol{\alpha})$ using the latent-count block and updates (a, d, p) using the generalized gamma block. The two blocks share the same conditional counts but contain disjoint current parameters, and accepted updates are checked against the observed penalized log-likelihood in Equation (14).

Algorithm 1 states the complete update and acceptance order. It treats a suboptimizer failure as a recorded failure and uses damping only to recover a finite block candidate that narrowly violates the observed-likelihood safeguard.

The negative-binomial model uses an expectation/conditional maximization either procedure.[12] Its conditional count is

$$\widehat{N}_i^{(k+1)} = \delta_i + \frac{(1 + \phi^{(k)}\delta_i)\theta_i^{(k)} S_{GG}^{(k)}(Y_i)}{1 + \phi^{(k)}\theta_i^{(k)} F_{GG}^{(k)}(Y_i)}. \tag{21}$$

Given these conditional counts, $(\beta_0, \boldsymbol{\alpha})$ and (a, d, p) are updated through Equations (18) and (19) with ϕ fixed at its current value. The dispersion is then updated by direct maximization of Equation (15), conditional on the other updated parameters. As $\phi \to 0$, Equation (21) converges to the Poisson conditional count in Equation (20), so the conditional and observed-data formulations share the same Poisson limit.

Algorithm 2 differs from Algorithm 1 only in the conditional count, predictor block, and observed-likelihood update of ϕ. This direct dispersion maximization is the ECME step.

Every accepted outer iteration must have finite parameters and may not reduce the relevant observed penalized log-likelihood beyond the declared monotonicity allowance. After the required minimum iterations, block convergence requires

$$\begin{aligned} \left|\ell^{(k+1)} - \ell^{(k)}\right| &\le \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}}\{1 + |\ell^{(k)}|\}, \\ \max_j |\eta_j^{(k+1)} - \eta_j^{(k)}| &\le \varepsilon_\eta, \end{aligned} \tag{22}$$

where η contains the free parameters on their optimization scales. The final designs use $\varepsilon_{\text{abs}} = 10^{-8}$, $\varepsilon_{\text{rel}} = 2 \times 10^{-8}$, and $\varepsilon_\eta = 10^{-5}$. The fitted object records objective values, block-optimizer status, maximum absolute parameter changes, iterations, runtime, convergence, and any failure code so that unsuccessful fits remain visible in subsequent summaries.

After the block iterations, a bounded joint optimization of the observed penalized log-likelihood refines the predictor, latency, and, when estimated, dispersion parameters. Convergence of this refinement requires finite parameters, a finite safeguarded objective, and a projected maximum score no larger than the declared tolerance. A nonfinite refinement or one that violates the likelihood safeguard is recorded as a refinement failure while preserving the last finite block state; a stopped optimizer is not relabelled as converged solely because its objective changed little. Boundary indicators for each positive parameter are retained because a numerically finite estimate at a bound can signal weak identification or model strain.

2.5 | Prediction, tuning, and evaluation

Prediction for an unseen covariate vector uses the training expansion $f(\boldsymbol{x}_{\text{new}}) = \widehat{\beta}_0 + K(\boldsymbol{x}_{\text{new}}, \boldsymbol{X}_{\text{train}})\widehat{\boldsymbol{\alpha}}$. Equations (9) and (12) then produce the individual cure fraction and population survival curve. Covariate preprocessing is estimated from the training data and applied unchanged to validation and test data.

Predictive evaluation uses common event-stratified outer splits with 70% of observations for training and 30% for testing. Kernel parameters, the penalty, model variants, and all preprocessing choices are selected inside the outer training set by a shared inner tuning protocol and comparable search budget. The same outer test observations and evaluation grid are used for every paired method comparison.

Let $\widehat{G}$ be the Kaplan–Meier estimate of the censoring survival distribution fitted from the outer training data, with censoring treated as the event. The weighting convention follows the prediction-error construction of Graf and colleagues.[13] For a test set of size m, the inverse-probability-of-censoring weighted Brier score is

$$\begin{aligned} \widehat{\text{BS}}(t) = \frac{1}{m}\sum_{i=1}^{m}\Bigg\{ &\frac{I(Y_i \le t, \delta_i = 1)}{\widehat{G}(Y_i-)}\widehat{S}_i(t)^2 \\ &+ \frac{I(Y_i > t)}{\widehat{G}(t)}[1 - \widehat{S}_i(t)]^2\Bigg\}. \end{aligned} \tag{23}$$

The integration horizon is selected from the outer training data and restricted to times with reliable censoring support. The integrated Brier score is the time average obtained by trapezoidal integration of Equation (23) over the frozen grid and division by the horizon length, without renormalizing by the realized sum of weights.

Evaluation summaries retain paired score differences, cure-fraction error in simulations, calibration, convergence, failures, and runtime. A common generalized gamma latency is used throughout the reported simulation so that the generating latency matches the fitted latency structure and predictor-shape contrasts are not confounded with unmodelled latency heterogeneity. Covariate-dependent latency was not evaluated in the reported scenarios and remains a future misspecification stress test.

3 | Simulation study

The primary simulation study was designed to separate predictor shape, latent-count dispersion, and sample size while keeping the generalized gamma latency correctly specified. Each scenario used two independent covariates, $x_1, x_2 \sim \text{Uniform}(-1, 1)$, and sample size $n \in \{300, 600\}$. Three cure-probability surfaces were crossed with Poisson counts and negative-binomial counts at dispersions $\phi \in \{0.25, 1.00\}$, giving 18 scenarios. The two negative-binomial settings represent modest and stronger count overdispersion, while the Poisson setting provides the nested limiting case. The bounded uniform domain deliberately differs from the historical source because it gives all three geometries a common domain for level-set and recovery comparisons. The source used standard-normal covariates and a covariate-dependent Weibull latency; retaining those choices would confound cure-predictor shape with unmodelled latency heterogeneity.

The true cure probability was generated from one of three fixed surfaces. Specifically, with $\text{expit}(v) = \{1+\exp(-v)\}^{-1}$, the designs

Algorithm 1 Penalized EM and bounded refinement for Poisson GG-KM

Require: Training data $(\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{\delta})$, kernel configuration, $\lambda \geq 0$, numerical tolerances, and iteration limits
Fit preprocessing to $\boldsymbol{X}$; construct $\boldsymbol{K}$; validate inputs and the symmetry/positive-semidefinite gate
Initialize $\eta^{(0)} = (\beta_0^{(0)}, \boldsymbol{\alpha}^{(0)}, \log a^{(0)}, \log d^{(0)}, \log p^{(0)})$
Evaluate and record $\ell_{\mathrm{P}}^{\mathrm{pen}}(\eta^{(0)})$
for $k = 0, \ldots, k_{\max} - 1$ do
  Compute $F_i^{(k)}$, $S_i^{(k)}$, $\theta_i^{(k)}$, and $\widehat{N}_i^{(k+1)} = \delta_i + \theta_i^{(k)} S_i^{(k)}$
  Maximize Equation (17) over $(\beta_0, \boldsymbol{\alpha})$
  Maximize Equation (19) over $(\log a, \log d, \log p)$
  Combine the block solutions into η_{cand} and evaluate $\ell_{\mathrm{P}}^{\mathrm{pen}}(\eta_{\mathrm{cand}})$
  if the candidate parameters or objective are nonfinite then
    Stop with a recorded nonfinite_objective failure
  end if
  if the candidate decreases the objective beyond the monotonicity tolerance then
    Search $\eta^{(k)} + 2^{-m}\{\eta_{\mathrm{cand}} - \eta^{(k)}\}$ over the bounded safeguard sequence
    if no finite safeguarded point is admissible then
      Stop with a recorded likelihood_decrease failure
    end if
  end if
  Accept the admissible candidate; record objective, optimizer states, parameter change, step fraction, and runtime
  if the objective-change and parameter-change criteria hold after the minimum iterations then
    Mark the block phase complete and leave the outer loop
  end if
end for
Unless the block phase failed, refine all free parameters jointly by bounded observed-likelihood optimization
Accept only a finite refinement that satisfies the objective safeguard; evaluate the projected maximum score
Return parameters, prediction state, full trace, boundary indicators, runtime, and the explicit convergence/failure code

were

$$\begin{aligned} p_c^{\mathrm{lin}}(\boldsymbol{x}) &= \mathrm{expit}(0.15 - 1.00x_1 + 0.75x_2), \\ p_c^{\mathrm{quad}}(\boldsymbol{x}) &= \mathrm{expit}(0.60 - 1.45x_1^2 - 1.05x_2^2 + 0.55x_1x_2), \\ p_c^{\mathrm{per}}(\boldsymbol{x}) &= \mathrm{expit}\{0.10 + 1.15\sin(\pi x_1) + 0.80\cos(\pi x_2)\}. \end{aligned} \tag{24}$$

For Poisson generation, the latent-count mean was $\theta(\boldsymbol{x}) = -\log p_c(\boldsymbol{x})$; for negative-binomial generation, it was $\theta(\boldsymbol{x}) = \{p_c(\boldsymbol{x})^{-\phi} - 1\}/\phi$. This mapping preserves the same cure-probability surface across count distributions, so comparisons do not confound dispersion with a different cure target.

Figure 3 shows the three surfaces on the common covariate square. The linear surface increases along one tilted gradient and has parallel straight level sets, so the log-linear predictor is correctly specified for it. The quadratic surface has closed elliptical level sets generated by the two squared terms and the interaction, and no additive predictor reproduces that geometry. The periodic surface reverses its gradient twice inside the domain and produces alternating regions of high and low cure probability, a shape an additive spline can follow one coordinate at a time but a degree-two map cannot. The three designs therefore separate correct linear specification, smooth nonadditivity and oscillation while holding the latency distribution fixed. They are not matched on prevalence: the mean of $p_c(\boldsymbol{x})$ over the square is 0.533 for the linear surface, 0.448 for the quadratic surface and 0.520 for the periodic surface, with attainable ranges $[0.168, 0.870]$, $[0.079, 0.646]$ and $[0.136, 0.886]$. Scores are consequently comparable between methods within a scenario and not between scenarios.

All latent-cause activation times followed the generalized gamma distribution with $(a, d, p) = (1.6, 1.3, 1.1)$. Independent censoring times were exponential with mean 3.5, and observed time was the minimum of the event and censoring times. The inverse population distribution was used to generate finite event times directly for susceptible individuals, while individuals in the limiting cure mass were assigned no event. Every replicate retained the true individual cure probability and population survival curve for evaluation.

Algorithm 3 states the implemented inverse-transform generator. Sampling from the improper population distribution avoids simulating a potentially large latent count while preserving its cure mass and finite-event distribution exactly.

Eight fitted methods shared the same generalized gamma latency: log-linear, additive cubic-spline, quadratic, and radial-basis GG-KM predictors, each with Poisson and negative-binomial latent counts. Log-linear and quadratic candidates used penalties $\lambda \in \{0, 0.1, 1, 10\}$, with the quadratic map containing all degree-two terms and interactions but no bias column. Spline candidates used three or five training-fold quantile knots with $\lambda \in \{0.1, 1\}$, and radial-basis candidates used $\gamma \in \{0.25, 1\}$ with $\lambda \in \{0.3, 3\}$. Each method therefore had four candidates and used the same two-fold inner tuning criterion and evaluation-grid size. The promotion-time support-vector method of Pal and Aselisewine

Algorithm 2 Penalized ECME and bounded refinement for negative-binomial GG-KM

Require: Training data $(\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{\delta})$, kernel configuration, $\lambda \geq 0$, optional fixed $\phi > 0$, tolerances, and iteration limits

Fit preprocessing, construct $\boldsymbol{K}$, apply the input and Gram-matrix checks, and initialize $(\beta_0, \boldsymbol{\alpha}, \log a, \log d, \log p, \log \phi)$

Evaluate and record $\ell_{\mathrm{NB}}^{\mathrm{pen}}$ at the initial state

for $k = 0, \ldots, k_{\max} - 1$ **do**

  Compute current $F_i^{(k)}$, $S_i^{(k)}$, $\theta_i^{(k)}$, and $\phi^{(k)}$

  Compute $\widehat{N}_i^{(k+1)}$ from Equation (21)

  Maximize Equation (18) over $(\beta_0, \boldsymbol{\alpha})$ with $\phi = \phi^{(k)}$

  Maximize Equation (19) over $(\log a, \log d, \log p)$

  if ϕ is estimated **then**

    Maximize Equation (15) over $\log \phi$ with the updated predictor and latency fixed

  else

    Retain the declared fixed dispersion

  end if

  Combine the conditional and ECME updates; evaluate the observed penalized likelihood

  if the candidate is nonfinite or violates the objective safeguard **then**

    Apply the same bounded interpolation and recorded-failure rule as Algorithm 1

  end if

  Accept the admissible candidate and record every block status, parameter change, step fraction, and runtime

  if both convergence criteria hold after the minimum iterations **then**

    Mark the block phase complete and leave the outer loop

  end if

end for

Unless the block phase failed, refine all free parameters jointly using the observed likelihood and analytical score

Accept only a finite safeguarded refinement; apply the projected-score convergence gate

Return parameters, prediction state, full trace, boundary indicators, runtime, and the explicit convergence/failure code

[Figure 3 near here]

FIGURE 3 | Frozen cure-probability surfaces used in the simulation design. Panels show the linear, quadratic, and periodic functions in Equation (24) over $(x_1, x_2) \in [-1, 1]^2$. All panels use the same $p_c(\boldsymbol{x})$ color scale; superimposed contour lines provide a color-independent encoding of the surface geometry. Contours are labelled at $p_c = 0.25$, 0.50 and 0.60, the same levels used in Figure 5, and each panel title reports the mean of $p_c(\boldsymbol{x})$ over the square.

was not included because the reproducibility audit found no public executable implementation at the article or arXiv records, and a new reimplementation would not provide a controlled software comparison.[4]

Each generated sample was split once into a common event-stratified 70% training set and 30% test set for all eight methods. Tuning, feature-map fitting, and preprocessing were confined to the training set, and the IPCW integration horizon was set from that set at the smaller of its 90th observed-time percentile and the limit at which estimated censoring survival remained at least 0.05. Primary test-set estimands were integrated Brier score, cure-probability bias, mean squared error, root mean squared error and mean absolute error, survival bias and mean squared error over subjects and grid times, and linear calibration intercepts and slopes against the known simulation truth. Failures, convergence, bound contact, selected hyperparameters, and runtime were retained rather than discarded.

A target-size diagnostic pilot used five replicates per pilot scenario and was not pooled with the final experiment. The largest observed between-replicate standard deviations were 0.0443 for integrated Brier score and 0.0703 for cure-probability root mean squared error. Requiring projected Monte Carlo standard errors no larger than 0.003 and 0.005, respectively, implied at least 218 replicates for the more restrictive criterion, so the final count was fixed at 250 replicates per scenario before final estimation. Final summaries report the realized Monte Carlo standard error for every metric and paired differences computed within the shared replicate and outer split.

3.1 | Simulation results

The primary final experiment contained 36,000 method rows from 250 replications in each of 18 scenarios. The largest realized Monte Carlo standard errors were 0.00183 for the IPCW integrated Brier score and 0.00243 for cure root mean squared error, below the pilot targets of 0.003 and 0.005. No fit failed. Of the 36,000 fits, 35,881 met the convergence criteria; method-specific totals ranged from 4,475 to 4,493 of 4,500, and all 119 nonconverged fits remained in the denominators. Mean per-fit runtime

Algorithm 3 Right-censored simulation from a promotion-time population distribution

Require: Scenario, sample size n, generalized gamma parameters (a, d, p), censoring scale, and deterministic replication seed
for $i = 1, \ldots, n$ do
  Draw $x_{i1}, x_{i2} \overset{\text{ind}}{\sim} \text{Uniform}(-1, 1)$ and evaluate the declared $p_c(\boldsymbol{x}_i)$
  Set $\theta_i = -\log p_c(\boldsymbol{x}_i)$ for Poisson counts, or $\theta_i = \{p_c(\boldsymbol{x}_i)^{-\phi} - 1\}/\phi$ for negative-binomial counts
  Draw $U_i \sim \text{Uniform}(0, 1)$ from the population event-time distribution
  if $U_i \geq 1 - p_c(\boldsymbol{x}_i)$ then
    Set $T_i = \infty$ (the draw falls in the cure mass)
  else
    Set $q_i = -\log(1 - U_i)/\theta_i$ for Poisson counts
    Set $q_i = \{(1 - U_i)^{-\phi} - 1\}/(\phi\theta_i)$ for negative-binomial counts
    Set $T_i = F_{GG}^{-1}(q_i; a, d, p)$
  end if
  Draw C_i from the declared independent censoring distribution
  Record $Y_i = \min(T_i, C_i)$, $\delta_i = I(T_i \leq C_i)$, $p_c(\boldsymbol{x}_i)$, and the true population survival grid
end for
Return the complete observed sample, generating truth, scenario identifier, and replication seed

ranged from 5.90 to 9.10 seconds across methods, with a maximum of 34.62 seconds. Estimated negative-binomial dispersion reached its lower bound in 10,711 of 18,000 negative-binomial fits and its upper bound in four, while 4,012 fits contacted a latency-parameter bound.

Table 2 reports the lowest mean IPCW integrated Brier score and cure root mean squared error in each scenario. For both metrics, a member of the log-linear family had the lowest mean in every linear scenario, a quadratic predictor in every quadratic scenario, and an additive spline in every periodic scenario. Thus the correctly aligned simple predictor, rather than GG-KM, attained the lowest scenario mean for each prespecified geometry. Within a predictor family, the largest difference between Poisson and negative-binomial fitted variants was 0.02 for integrated Brier score and 0.01 for cure root mean squared error.

Figure 4 shows the six count-matched contrasts of GG-KM against log-linear, spline, and quadratic predictors. On the linear surface, GG-KM's integrated Brier difference from the log-linear model ranged from +0.00182 to +0.00363, and its cure root mean squared error difference ranged from +0.01971 to +0.02517; all 12 count-specific contrasts were resolved at 1.96 Monte Carlo standard errors. The quadratic predictor also had lower mean error than GG-KM in most linear comparisons. GG-KM had lower cure root mean squared error than the spline throughout the linear scenarios, illustrating that pairwise ordering depends on the comparator as well as the surface.

On the quadratic surface, GG-KM improved on the log-linear and additive-spline predictors but not on the correctly aligned quadratic comparator. In the Poisson-generated $n = 600$ cell, GG-KM Poisson minus quadratic Poisson was +0.00155 (Monte Carlo standard error 0.00027) for integrated Brier score and +0.01496 (0.00106) for cure root mean squared error. In the same cell, its differences from log-linear Poisson were −0.00978 (0.00059) and −0.06112 (0.00125), respectively. On the periodic surface, GG-KM improved on the log-linear and quadratic predictors but trailed the spline. The corresponding $n = 600$ Poisson differences from spline Poisson were +0.00099 (0.00036) and +0.00863 (0.00115).

GG-KM calibration against the known simulation truth remained close to identity across the declared scenarios: cure calibration intercepts ranged from −0.024 to 0.060 with slopes from 0.891 to 1.066, and survival calibration intercepts ranged from 0.008 to 0.058 with slopes from 0.913 to 0.996.

Figure 5 maps where in the covariate square each predictor loses accuracy. Every panel fills the 41×41 evaluation grid with the pointwise root mean squared error of the predicted cure probability against the generating surface, averaged over the 250 Poisson replications at $n = 600$, on a scale shared by all twelve panels. Solid lines repeat the true level sets at $p_c = 0.25$, 0.50 and 0.60, and the dashed line is the mean estimated $p_c = 0.50$ level, which separates an error of magnitude from a displacement of shape. Every grid point retained all 250 replications. The panel means reproduce the scalar comparison: on the linear surface the log-linear predictor attains 0.0498 against 0.0756 for GG-KM, on the quadratic surface the quadratic predictor attains 0.0631 against 0.0787, and on the periodic surface the spline attains 0.0856 against 0.0941. The spatial pattern adds what those six numbers omit. A misaligned predictor does not spread its error evenly. The log-linear predictor reaches a pointwise error of 0.374 on the quadratic surface and 0.433 on the periodic surface, concentrated where the true surface curves most, and its estimated $p_c = 0.50$ level is absent on the quadratic surface because its fitted range collapses to $[0.443, 0.448]$. The quadratic predictor reaches 0.349 on the periodic surface. The radial-basis GG-KM predictor does not exceed 0.2063 on any surface and its estimated $p_c = 0.50$ level follows the true level in every panel. Figure 5 is therefore the pointwise counterpart of the regret summary in Table 3: the kernel predictor is never the most accurate on a surface, and it is also never the predictor that fails on one.

The complete mean IPCW integrated Brier scores for all eight methods, with their Monte Carlo standard errors, appear in the Supporting Information. They confirm the geometry-specific ordering in Table 2 and retain both fitted count laws for every predictor.

[Figure 4 near here]

FIGURE 4 | Count-matched paired predictive differences of radial-basis GG-KM against the three comparator predictors. The top row shows mean IPCW integrated Brier differences and the bottom row cure root mean squared error differences, with columns for linear, quadratic, and periodic cure surfaces. Each panel contains exactly six contrasts: GG-KM Poisson minus log-linear, spline, and quadratic Poisson, and GG-KM negative binomial minus the corresponding three negative-binomial methods. Error bars span ±1.96 paired Monte Carlo standard errors. A horizontal line marks zero; negative values favor GG-KM.

[Figure 5 near here]

FIGURE 5 | Pointwise accuracy of the three primary cure surfaces at $n = 600$ under Poisson generation. Rows show the linear, quadratic and periodic generating surfaces; columns show the log-linear, additive-spline, quadratic and radial-basis GG-KM predictors. The filled field is the root mean squared error of the predicted cure probability at each of the 41×41 grid points, taken across all valid replications, on one scale shared by every panel and truncated above 0.30. Solid labelled lines are the true cure levels 0.25, 0.50 and 0.60; the dashed line is the mean estimated 0.50 level and is annotated as absent when the fitted range does not contain it. The value printed in each panel is the mean of the displayed field. Failed fits remain in n_{total} and reduce the pointwise n_{valid} rather than disappearing.

3.2 | Regret within the evaluated set

Table 3 limits regret to the eight fitted methods and 18 scenarios. Negative-binomial GG-KM had the smallest maximum paired integrated-Brier regret, 0.00347 with a 95% Monte Carlo half-width of 0.00098; Poisson GG-KM was close at 0.00364 with a half-width of 0.00099, so the intervals overlap. Its equally weighted mean regret was 0.00189 (half-width 0.00021). For cure root mean squared error, negative-binomial GG-KM also had the smallest maximum regret, 0.02516 (half-width 0.00238), and mean regret 0.01606 (half-width 0.00061). This low maximum regret did not imply the lowest descriptive mean rank: the two quadratic methods averaged rank 3.50 for integrated Brier score, compared with 4.00 for negative-binomial GG-KM.

3.3 | Higher-dimensional interaction surfaces

The four-covariate experiment is a secondary mechanism study developed after the primary analysis. It probes interaction structures that an additive spline cannot represent, but it is neither independent confirmation nor evidence of a general kernel advantage. Covariates satisfy $x_1, \ldots, x_4 \sim \text{Uniform}(-1, 1)$, and the first target is the *interaction* surface

$$p_c(\boldsymbol{x}) = \text{expit}\big(C_{\text{int}} + 1.60\, x_1 x_2 - 1.30\, x_3 x_4 + 0.70\, x_1 - 0.50\, x_4\big), \tag{25}$$

and the *radial* surface is

$$p_c(\boldsymbol{x}) = \text{expit}\big(C_{\text{rad}} + 3.00 \exp\{-0.80(x_1^2 + x_2^2 + x_3^2 + x_4^2)\}\big). \tag{26}$$

Intercepts C_{int} and C_{rad} were calibrated by bisection on a fixed sample of 200 000 observations (seed 20260901) so that each surface yields a mean cure fraction of 0.45. The latency remained the common generalized gamma with $(a, d, p) = (1.6, 1.3, 1.1)$ and no covariate-dependent scale; censoring was exponential with mean 3.5. The same eight methods were compared under Poisson, negative-binomial ($\phi = 0.25$), and negative-binomial ($\phi = 1.00$) generating laws at $n = 300$ and $n = 600$, giving 12 scenarios. The replication count was chosen from a separate target-size pilot using the same precision rule as the primary design, and every declared cell remains in the analysis regardless of its ordering.
Table 4 and Figure 6 report the directed comparisons with paired Monte Carlo uncertainty. Their interpretation is limited to the evaluated interaction and radial mechanisms, and it retains every declared cell, including comparisons that do not favor GG-KM.
The complete per-scenario means with Monte Carlo standard errors, together with the numerical diagnostics, are reported in the Supporting Information.

4 | Melanoma application

The empirical application used the malignant melanoma cohort distributed as the `melanoma` data set in version 1.3-32 of the `boot` R package and documented in the original counting-process source.[14,15] The cohort contains 205 patients operated on in Odense, Denmark between 1962 and 1977, with follow-up recorded through 1977. There were 57 deaths from melanoma, 14 deaths from other causes, and 134 patients alive at the end of follow-up. The analysis treated death from melanoma as the event of interest and censored both administrative survivors and deaths from other causes, so the fitted model is cause specific in its event definition and is not a competing-risks analysis; $p_c(\boldsymbol{x})$ is the probability of no melanoma death from the modeled process, not an all-cause survival probability. The predictor set comprised sex, age at operation, calendar year of operation, the natural logarithm of tumour thickness in millimetres, and ulceration status. Follow-up time was converted from days to years, and no observations or covariate values were missing. Continuous and binary predictors passed through the same training-fold standardization step so that no test-set information entered preprocessing. The versioned analysis copy was confirmed row for row against `boot::melanoma` and audited against its recorded SHA-256 checksum before estimation.
Predictive performance was assessed in 50 event-stratified outer holdouts, each assigning 143 patients to training and 62 to testing. The same split indices were used for every method, and all tuning choices were made within three inner training folds using the IPCW integrated Brier score and the horizon rule defined in Section 2.5. The comparison crossed log-linear, additive cubic-spline, quadratic, and radial-basis GG-KM predictors with Poisson and negative-binomial latent-count laws, giving eight methods. Penalty, feature-map, spline, and kernel choices were restricted to four prespecified candidates per method, and no test outcome contributed to preprocessing or model selection.
Table 1 reports split-level means and variation. The log-linear Poisson model had the lowest mean IPCW integrated Brier score,

[Figure 6 near here]

FIGURE 6 | Paired predictive differences of the radial-basis GG-KM predictor against each count-matched comparator on the interaction and radial cure surfaces. The top row shows paired mean IPCW integrated Brier differences and the bottom row paired mean cure root mean squared error differences. Each panel plots six contrasts across the six generating-law and sample-size cells, with error bars spanning ±1.96 Monte Carlo standard errors.

0.1384 with split standard deviation 0.0186, followed by the log-linear negative-binomial model at 0.1392 with standard deviation 0.0190. The two GG-KM count models were nearly indistinguishable at 0.1415 with standard deviation 0.0151, and the paired GG-KM Poisson minus log-linear Poisson difference averaged 0.0030 with empirical split quantiles from −0.0143 to 0.0214, so the paired difference is not resolved away from zero across splits. The spline and quadratic models had higher mean scores in this application, while all 400 fits returned finite predictions and 391 met the explicit convergence criteria. Convergence ranged from 45 of 50 splits for GG-KM Poisson to 50 of 50 splits for the quadratic and spline models.

Figure 7 places the paired score distributions beside descriptive survival calibration. The comparison of split-averaged predicted survival with the Kaplan–Meier curve is a calibration check on the population survival function; it does not identify individual cure status. Repeated holdouts are dependent because the same patients recur across splits, so empirical quantiles describe variation across training sets rather than independent-sample confidence intervals. Similar mean scores do not establish an approximately linear latent-cause structure, and a smaller split standard deviation alone does not prove greater predictive stability.

Mean predicted cure fractions ranged from 0.577 for negative-binomial GG-KM to 0.648 for the negative-binomial spline model, with the Poisson GG-KM estimate equal to 0.577. These values are model-based extrapolations because individual cure status is unobserved and follow-up is finite, and the Kaplan–Meier comparison cannot identify a cure fraction. Across-split prediction summaries were retained for every individual to expose variation caused by repeated training samples rather than to create surrogate cure labels. The application consequently evaluates reproducible out-of-sample prediction and numerical behavior while leaving biological interpretation of individual cure probabilities outside its evidential scope.

5 | Discussion

The generalized gamma kernel-machine promotion-time cure model extends the promotion-time framework by placing the latent-cause log-intensity in a reproducing-kernel Hilbert space while retaining a flexible generalized gamma latency and both Poisson and negative-binomial latent counts. The primary and secondary simulations and the melanoma application evaluate this construction against simpler predictors under a shared protocol. Their interpretation is limited to the evaluated designs. Kernel-family robustness was not evaluated and is left as future work.

The principal simulation pattern was consistent across the two latent-count laws. On the quadratic surface GG-KM lowered the paired IPCW integrated Brier score against the log-linear model by about 0.007 to 0.012 and the paired cure root mean squared error by about 0.04 to 0.07, with a smaller but resolved advantage over the spline. On the periodic surface GG-KM improved substantially on the log-linear model yet trailed the spline by a small margin, and on the linear surface it paid a small resolved penalty against the correctly specified log-linear model. The count law was a secondary factor, since Poisson and negative-binomial variants within a predictor family differed by less than 0.02 in integrated Brier score. The melanoma cohort produced overlapping score distributions in which the log-linear Poisson model had the lowest mean integrated Brier score and GG-KM was competitive with the smallest split-to-split variation, so the empirical application reinforces the simulation message that no single predictor is uniformly best.

Several features of the cure-model setting limit how far these results can be interpreted. Cure fractions are only weakly identified from right-censored data with finite follow-up, because a long plateau of censored survivors is consistent with a range of cure probabilities. The melanoma follow-up ends in 1977 and the estimated cure fractions between 0.577 and 0.648 are model-based extrapolations, not quantities read directly from the data. The comparison of split-averaged predicted survival with the Kaplan-Meier curve is a calibration check on the population survival function and does not identify which individuals are cured. The analysis therefore makes no claim to identify individual cure status, and simulation calibration slopes near identity do not change that limitation.

Kernel choice and its hyperparameters shape the estimated intensity because they define which subjects are similar and how rapidly the fitted surface may change. The reported results used radial-basis kernels selected from a small prespecified grid by leakage-free inner tuning. Bandwidths that are too small can track idiosyncratic training variation, whereas overly large bandwidths approach a much smoother response; both effects depend on the training-fold scaling used to measure distance. A different kernel family or bandwidth range could therefore shift the balance between GG-KM and the spline, particularly on surfaces whose curvature does not match radial-basis smoothness, as the periodic surface illustrates. Kernel and penalty selection are model-selection steps, and uncertainty from that selection is not represented by the fitted curve alone. The comparisons fixed a common tuning budget and evaluation protocol for all methods, so the reported differences describe performance at a shared and modest search intensity rather than each method's best attainable configuration.

The kernel-machine estimator also carries a computational cost that grows with sample size. An $n_{\text{train}} \times n_{\text{train}}$ Gram matrix requires quadratic storage, and dense factorization can require cubic time; nested tuning repeats those operations across candidate kernels and inner folds. Reusing a Gram matrix within a candidate and retained eigenspace avoids unnecessary work, but matrices cannot be shared across folds whose preprocessing and training subjects differ. These costs were modest at the sample sizes studied, with mean fit times of a few seconds, but those timings do not establish scalability to substantially larger cohorts. Low-rank, Nyström, random-feature, or inducing-point approximations could reduce the burden, although each changes the effective penalty and

[Figure 7 near here]

FIGURE 7 | Predictive assessment in 50 paired repeated holdouts of the melanoma cohort. Panel A compares the full-cohort Kaplan–Meier curve with split-averaged out-of-sample survival predictions over the common supported horizon. Panel B shows split-level IPCW integrated Brier scores, with each point representing one outer split and lower values indicating better prediction. Line type, marker or box encoding, and labels distinguish all eight methods without relying on color alone. Panel A shows mean curves without an uncertainty band; the dependence caveat for repeated holdouts is stated in the text.

out-of-sample predictor and would require separate numerical validation. Sample size affected the strength of the conclusions as well as runtime: increasing the sample from 300 to 600 reduced Monte Carlo uncertainty and resolved the quadratic and periodic contrasts more clearly.

Censoring and the event definition further bound the interpretation. The evaluation used inverse-probability-of-censoring weighting with a horizon confined to the region of reliable censoring support, which controls bias from dependent censoring only to the extent that the censoring model is adequate. In the melanoma cohort, deaths from other causes were treated as censored, so the fitted model is cause specific in its event definition and is not a competing-risks analysis. Within that latent cause-specific process, $p_c(\boldsymbol{x})$ describes the probability of no melanoma death from the modeled process; it is neither an all-cause survival probability nor a competing-risks cumulative incidence. The interpretation relies on independent censoring of other-cause deaths. The negative-binomial dispersion was only weakly identified from cure and latency information: under Poisson generation the estimated dispersion correctly stayed near zero, but under true dispersion of 1.00 it was systematically underestimated, ranging from 0.053 to 0.239. This weak identifiability explains the frequent contact of the dispersion estimate with its lower bound and cautions against interpreting the fitted dispersion as a precise measure of overdispersion.

The melanoma application is observational, and no causal interpretation is attached to the covariate effects or to the predicted cure fractions. The reported associations describe out-of-sample predictive behavior under a fixed analysis protocol, not the effect of intervening on any covariate.

Future work could extend the estimator in directions that were deliberately left outside the present scope. A controlled comparison against reproducible support-vector or deep cure models would sharpen the positioning of kernel machines among nonlinear approaches, but a fair comparison requires public implementations of the same promotion-time model with common generalized gamma latency, which were not available, so such comparators are noted as open rather than completed. Covariate-dependent latency, low-rank Gram approximations for larger cohorts, richer kernel families, and joint modeling of the censoring mechanism are natural extensions. The current contribution is the validated combination of an RKHS latent-cause intensity, generalized gamma latency, Poisson and negative-binomial latent counts, penalized EM and ECME estimation, and reproducible leakage-free evaluation with paired Monte Carlo uncertainty.

6 | Declarations

Data and code availability. The malignant melanoma data are distributed as `boot::melanoma` in version 1.3-32 of the `boot` R package, whose official CRAN metadata report an Unlimited license, and are cited to the package documentation and original source. The reproducible article workspace contains the row-for-row verified analysis copy, its SHA-256 checksum, executable Python code, frozen design records, session information, checkpoints, and machine-readable outputs for every table and figure. **Author confirmation required:** deposit the final data and code package in a public archive, insert its persistent identifier, and verify that the archived files reproduce the submitted version.

Ethics and consent. This study reanalyzes a publicly distributed historical cohort and does not report newly collected participant data. The available thesis and package documentation do not establish whether a new institutional determination is required for this secondary analysis. **Author confirmation required:** provide the journal-appropriate ethics approval, exemption, or not-applicable statement and confirm whether a consent statement is required.

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TABLE 1 | Predictive performance in 50 paired repeated holdouts of the melanoma cohort. Values for the IPCW integrated Brier score and mean predicted cure fraction are split means with split standard deviations in parentheses. Runtime is the mean with the maximum in parentheses, and lower integrated Brier scores indicate better prediction over the common evaluation horizon.

Predictor	Count law	Converged	IPCW IBS	Mean cure prediction	Runtime (s)
Log-linear	Poisson	49/50	0.1384 (0.0186)	0.604 (0.057)	7.08 (14.79)
Log-linear	Negative binomial	49/50	0.1392 (0.0190)	0.619 (0.054)	9.23 (14.81)
Spline	Poisson	50/50	0.1616 (0.0279)	0.616 (0.060)	9.10 (16.31)
Spline	Negative binomial	50/50	0.1518 (0.0226)	0.648 (0.042)	10.79 (16.67)
Quadratic	Poisson	50/50	0.1444 (0.0229)	0.615 (0.058)	6.69 (12.36)
Quadratic	Negative binomial	50/50	0.1401 (0.0199)	0.637 (0.046)	8.27 (13.52)
GG-KM	Poisson	45/50	0.1415 (0.0151)	0.577 (0.053)	8.01 (15.67)
GG-KM	Negative binomial	48/50	0.1415 (0.0151)	0.577 (0.053)	8.28 (14.89)

TABLE 2 | Lowest mean prediction errors in each simulation scenario. Parenthesized values are Monte Carlo standard errors, and the full eight-method summaries are reported in the Supporting Information. The lowest observed mean is a descriptive comparison, not a claim of superiority when paired uncertainty includes zero. Lin, Spl, Quad, GG, P, and NB denote log-linear, spline, quadratic, GG-KM, Poisson, and negative binomial, respectively.

True surface	Generating law	n	Lowest mean IPCW IBS	Lowest mean cure RMSE
Linear	P	300	Lin-P: 0.1937 (0.0014)	Lin-P: 0.0659 (0.0018)
Linear	NB(0.25)	300	Lin-NB: 0.1946 (0.0014)	Lin-P: 0.0635 (0.0016)
Linear	NB(1.00)	300	Lin-NB: 0.1989 (0.0015)	Lin-NB: 0.0627 (0.0016)
Linear	P	600	Lin-P: 0.1929 (0.0009)	Lin-P: 0.0465 (0.0011)
Linear	NB(0.25)	600	Lin-P: 0.1960 (0.0010)	Lin-P: 0.0463 (0.0011)
Linear	NB(1.00)	600	Lin-P: 0.1997 (0.0011)	Lin-NB: 0.0430 (0.0012)
Quadratic	P	300	Quad-P: 0.2096 (0.0013)	Quad-NB: 0.0857 (0.0018)
Quadratic	NB(0.25)	300	Quad-P: 0.2093 (0.0015)	Quad-NB: 0.0830 (0.0015)
Quadratic	NB(1.00)	300	Quad-NB: 0.2151 (0.0014)	Quad-NB: 0.0803 (0.0016)
Quadratic	P	600	Quad-P: 0.2070 (0.0009)	Quad-P: 0.0603 (0.0012)
Quadratic	NB(0.25)	600	Quad-NB: 0.2087 (0.0009)	Quad-NB: 0.0606 (0.0012)
Quadratic	NB(1.00)	600	Quad-NB: 0.2133 (0.0010)	Quad-NB: 0.0580 (0.0012)
Periodic	P	300	Spl-NB: 0.1867 (0.0015)	Spl-NB: 0.1120 (0.0020)
Periodic	NB(0.25)	300	Spl-P: 0.1922 (0.0015)	Spl-NB: 0.1110 (0.0019)
Periodic	NB(1.00)	300	Spl-NB: 0.1939 (0.0018)	Spl-NB: 0.1021 (0.0017)
Periodic	P	600	Spl-P: 0.1851 (0.0011)	Spl-NB: 0.0823 (0.0012)
Periodic	NB(0.25)	600	Spl-NB: 0.1870 (0.0012)	Spl-NB: 0.0794 (0.0011)
Periodic	NB(1.00)	600	Spl-NB: 0.1866 (0.0012)	Spl-NB: 0.0742 (0.0010)

TABLE 3 | Paired regret and descriptive rank summaries over the 18 primary simulation scenarios. Within each scenario, the reference is the method with the lowest finite mean. Regret uses only replication pairs for which both methods are finite. The worst regret identifies its scenario and reports a 95% Monte Carlo half-width; mean regret weights scenarios equally and propagates their Monte Carlo standard errors. Mean rank is based on scenario means and is descriptive.

Metric	Method	Worst paired regret ($\pm$95%; scenario)	Mean paired regret ($\pm$95%)	Mean rank
IPCW integrated Brier score	Log-linear Poisson	0.0200 ± 0.0015 (Periodic, NB(1.00), $n = 600$)	0.0088 ± 0.0003	5.72
IPCW integrated Brier score	Log-linear Negative binomial	0.0200 ± 0.0015 (Periodic, NB(1.00), $n = 600$)	0.0087 ± 0.0003	5.28
IPCW integrated Brier score	Spline Poisson	0.0052 ± 0.0013 (Linear, P, $n = 300$)	0.0025 ± 0.0002	5.00
IPCW integrated Brier score	Spline Negative binomial	0.0054 ± 0.0013 (Linear, P, $n = 300$)	0.0023 ± 0.0002	4.44
IPCW integrated Brier score	Quadratic Poisson	0.0092 ± 0.0011 (Periodic, NB(1.00), $n = 600$)	0.0029 ± 0.0002	3.50
IPCW integrated Brier score	Quadratic Negative binomial	0.0092 ± 0.0011 (Periodic, NB(1.00), $n = 600$)	0.0028 ± 0.0002	3.50
IPCW integrated Brier score	GG-KM Poisson	0.0036 ± 0.0010 (Linear, NB(1.00), $n = 300$)	0.0019 ± 0.0002	4.56
IPCW integrated Brier score	GG-KM Negative binomial	0.0035 ± 0.0010 (Linear, NB(1.00), $n = 300$)	0.0019 ± 0.0002	4.00
Cure RMSE	Log-linear Poisson	0.0931 ± 0.0020 (Periodic, NB(1.00), $n = 600$)	0.0488 ± 0.0005	5.50
Cure RMSE	Log-linear Negative binomial	0.0931 ± 0.0020 (Periodic, NB(1.00), $n = 600$)	0.0487 ± 0.0005	5.50
Cure RMSE	Spline Poisson	0.0386 ± 0.0030 (Linear, NB(0.25), $n = 300$)	0.0207 ± 0.0005	5.33
Cure RMSE	Spline Negative binomial	0.0382 ± 0.0036 (Linear, P, $n = 300$)	0.0182 ± 0.0005	4.33
Cure RMSE	Quadratic Poisson	0.0534 ± 0.0017 (Periodic, NB(1.00), $n = 600$)	0.0203 ± 0.0004	3.61
Cure RMSE	Quadratic Negative binomial	0.0529 ± 0.0017 (Periodic, NB(1.00), $n = 600$)	0.0200 ± 0.0004	3.39
Cure RMSE	GG-KM Poisson	0.0252 ± 0.0024 (Linear, P, $n = 600$)	0.0162 ± 0.0006	4.39
Cure RMSE	GG-KM Negative binomial	0.0252 ± 0.0024 (Linear, P, $n = 600$)	0.0161 ± 0.0006	3.94

TABLE 4 | Lowest mean prediction errors in each high-dimensional simulation scenario. Parenthesized values are Monte Carlo standard errors, and the full eight-method summaries are reported in the Supporting Information. Int and Rad denote Interaction and Radial surfaces.

True surface	Generating law	n	Lowest mean IPCW IBS	Lowest mean cure RMSE
Interaction	P	300	quadratic-poisson: 0.2070 (0.0020)	quadratic-poisson: 0.1289 (0.0021)
Interaction	NB(0.25)	300	quadratic-negative-binomial: 0.2082 (0.0022)	quadratic-negative-binomial: 0.1305 (0.0026)
Interaction	NB(1.00)	300	quadratic-negative-binomial: 0.2075 (0.0023)	quadratic-negative-binomial: 0.1217 (0.0020)
Interaction	P	600	quadratic-negative-binomial: 0.2017 (0.0013)	quadratic-negative-binomial: 0.0985 (0.0016)
Interaction	NB(0.25)	600	quadratic-negative-binomial: 0.2038 (0.0014)	quadratic-negative-binomial: 0.0945 (0.0015)
Interaction	NB(1.00)	600	quadratic-negative-binomial: 0.1998 (0.0015)	quadratic-negative-binomial: 0.0865 (0.0013)
Radial	P	300	ggkm-negative-binomial: 0.2180 (0.0021)	ggkm-negative-binomial: 0.1224 (0.0023)
Radial	NB(0.25)	300	ggkm-negative-binomial: 0.2200 (0.0018)	ggkm-negative-binomial: 0.1170 (0.0020)
Radial	NB(1.00)	300	ggkm-poisson: 0.2216 (0.0022)	ggkm-poisson: 0.1138 (0.0021)
Radial	P	600	ggkm-negative-binomial: 0.2126 (0.0013)	ggkm-negative-binomial: 0.1010 (0.0018)
Radial	NB(0.25)	600	ggkm-negative-binomial: 0.2139 (0.0013)	ggkm-negative-binomial: 0.0996 (0.0016)
Radial	NB(1.00)	600	ggkm-negative-binomial: 0.2208 (0.0014)	ggkm-negative-binomial: 0.0973 (0.0015)